Inter-VLAN Routing Lab

1) **Title**: Inter-VLAN Routing using Router-on-a-Stick in Cisco Packet Tracer.

2) **Aim / Objectives :**

To configure Inter-VLAN Routing using Router-on-a-Stick and enable communication between devices in different VLANs.

3) **Devices Used :**

- 1 Switch (2960)
- 1 Router (2911)
- 4 PCs
- Copper Straight-through Cables

4) **Topology :**

```
PC0 ----|
        |
PC1 ----|       Switch0 -------- Router0
        |
PC2 ----|
        |
PC3 ----|
```

5) **VLAN Configuration :**

VLAN ID	VLAN Name	Devices
10	HR	PC0, PC1
20	FINANCE	PC2, PC3

6) IP Addressing Table:

VLAN 10

Devices	IP Address	Subnet mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1

VLAN 20

Devices	IP Address	Subnet mask	Default Gateway
PC0	192.168.20.2	255.255.255.0	192.168.20.1
PC1	192.168.20.3	255.255.255.0	192.168.20.1

7) Port Assignment:

Switch Port	Device	VLAN
Fa0/1	PC0	VLAN 10
Fa0/2	PC1	VLAN 10
Fa0/3	PC2	VLAN 20
Fa0/4	PC3	VLAN 20
Fa0/24	Router	Trunk

8) Procedure / Steps :

Step 1: Create VLANs

Open Switch CLI and enter:

```
enable
configure terminal

vlan 10
name HR
exit
```

```
vlan 20
 name FINANCE
 exit
```

Step 2: Assign Ports to VLAN 10

```
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
exit
```

Step 3: Assign Ports to VLAN 20

```
interface range fa0/3 - 4
switchport mode access
switchport access vlan 20
exit
```

Step 4: Configure Trunk Port

```
interface fa0/24
switchport mode trunk
exit
```

Step 5: Configure Router Interface

```
enable
configure terminal

interface g0/0
no shutdown
exit
```

Step 6: Configure Subinterface for VLAN 10

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

Step 7: Configure Subinterface for VLAN 20

```
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

Step 8: Configure IP Addresses on PCs

Configure the following:

PC0

```
IP Address: 192.168.10.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1
```

PC1

```
IP Address: 192.168.10.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1
```

PC2

```
IP Address: 192.168.20.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1
```

PC3

```
IP Address: 192.168.20.3
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1
```

Verification

Verify VLAN Configuration

```
show vlan brief
```

Verify Trunk Port

```
show interfaces trunk
```

Verify Router Interfaces

```
show ip interface brief
```

Expected Output:

```
GigabitEthernet0/0.10    up    up
GigabitEthernet0/0.20    up    up
```

Connectivity Test

Same VLAN Communication

From PC0:

`ping 192.168.10.3`

Result: Successful

Inter-VLAN Communication

From PC0:

`ping 192.168.20.2`

Result: Successful

From PC1:

`ping 192.168.20.3`

Result: Successful

9) Result :

Inter-VLAN Routing was successfully configured using Router-on-a-Stick. Devices belonging to different VLANs were able to communicate through the router.

10) Concepts Learned :

- VLAN Creation
- VLAN Segmentation
- Trunk Port Configuration
- Router-on-a Stick Configuration
- Subinterfaces
- 802. 1Q Encapsulation
- Inter-VLAN Communication

11) Cybersecurity Relevance :

- Network Segmentation.
- Controlled Communication Between Departments.
- Reduced Attack Surface
- Foundation for Access Control Lists (ACLs)
- Improved Network Security and Management.

12) Screenshot :